

CST8257 Web Application Development

Project

This project is only for those who have extra capacities. You can work by yourself on this project to practice more on PHP web development. It will not affect your course's final grade whether you choose to do this project or not.

Objective

Apply knowledge learnt in this course to create a social media network website for picture album management and sharing.

Due Date

See Brightspace post for the due of this project. To earn 10 points, you are required:

1. Complete this project in a group of 2 – 3 students as required.
2. Submit your project work to the Brightspace before the due date. Only one submission is required for each group.
3. Demonstrate your project work during the lab session **before** the due date.

Requirements

This is the comprehensive project of this course. Compared with the other labs, the amount of workload evolved in this project is significant. You should reuse your work in lab 5 and lab 6 as much as possible. Your final deliverables must be at **commercial-grade**. To be specific:

1. The layout and the styles of the pages may not necessary be the same as the screen captures shown in this document, but they should be well designed and styled like being developed by professional web developers.
2. The website is secure. At minimum, it is protected from password stealing and SQL injection.

Database

1. Create a database CST8257 and create the following tables in the database **CST8257**:

User

UserId varchar(16) NOT NULL PRIMARY KEY,
Name varchar(256) NOT NULL,
Phone varchar(16),
Password varchar(256)

Accessibility

Accessibility_Code varchar(16) NOT NULL PRIMARY KEY,
Description varchar(127) NOT NULL

Album

Album_Id int(11) PRIMARY KEY AUTO_INCREMENT,
Title varchar(256) NOT NULL,
Description varchar(3000),
Date_Updated Date NOT NULL,
Owner_Id varchar(16) NOT NULL,
Accessibility_Code varchar(16) NOT NULL,
FOREIGN KEY (Owner_Id) REFERENCES **User** (UserId),
FOREIGN KEY (Accessibility_Code) REFERENCES **Accessibility** (Accessibility_Code)

Picture

Picture_Id, int(11) NOT NULL PRIMARY KEY AUTO_INCREMENT,
Album_Id, int(11), NOT NULL,
FileName, varchar(255) NOT NULL,
Title, varchar(256) NOT NULL,
Description, varchar(3000),
Date_Added date NOT NULL,
FOREIGN KEY (Album_Id) REFERENCES **Album** (Album_Id)

Comment

Comment_Id int(11) NOT NULL PRIMARY KEY AUTO_INCREMENT,
Author_Id varchar(16) NOT NULL,
Picture_Id int(11) NOT NULL,
Comment_Text varchar(3000) NOT NULL,
Date timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (Author_Id) REFERENCES **User** (UserId),
FOREIGN KEY (Picture_Id) REFERENCES **Picture** (Picture_Id)

FriendshipStatus

Status_Code varchar(16) NOT NULL PRIMARY KEY,
Description varchar(128) NOT NULL,

Friendship

Friend_RequesterId varchar(16) NOT NULL PRIMARY KEY,
Friend_RequesteeId varchar(16) NOT NULL PRIMARY KEY,
Status varchar(16) NOT NULL,
FOREIGN KEY (Friend_RequesterId) REFERENCES **User** (UserId),
FOREIGN KEY (Friend_RequesteeId) REFERENCES **User** (UserId),
FOREIGN KEY (Status) REFERENCES **FriendshipStatus** (Status_Code)

2. Run SQL statements in file **Script for Populating Project DB tables.sql** (downloadable from the Brightspace) against database CST8257 to populate **Accessibility** table and **FriendshipStatus** table.
3. The follow is the relational data model of this website. Study this relational data model to see how you can derive entity classes from the database tables in the model.



All pages on this website have the following header.

localhost:8080/CST8257Project/Index.php

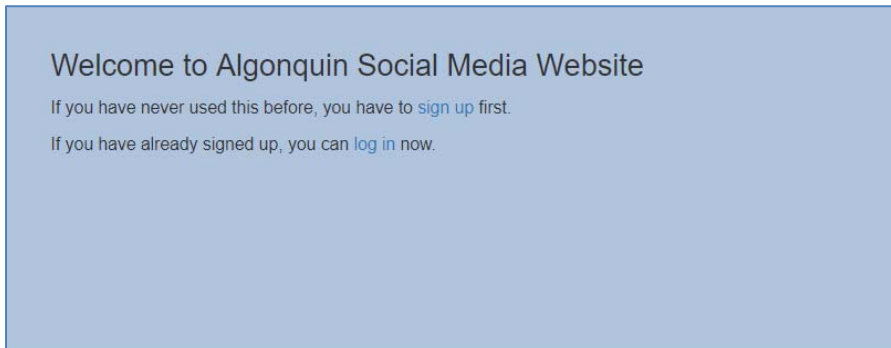
Home My Friends My Albums My Pictures Upload Pictures Log In

- Home** – link to Index.php page
- My Friends** – link to MyFriends.php page
- My Albums** – link to MyAlbums.php page
- My Pictures** – link to MyPictures.php page
- Upload Pictures** – link to UploadPictures.php page
- Log In / Log Out** – link to Login.php if the user is not logged in yet, or Logout.php page if the user has already logged in.

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2. Index.php

This page is the landing (default) page of this website.



3. NewUser.php

The same as in lab 5, this page is for a user to sign up with the system. The same validation rules as in lab 5 apply as well. **Passwords cannot be saved in the User table as plain text. They must be hashed.**

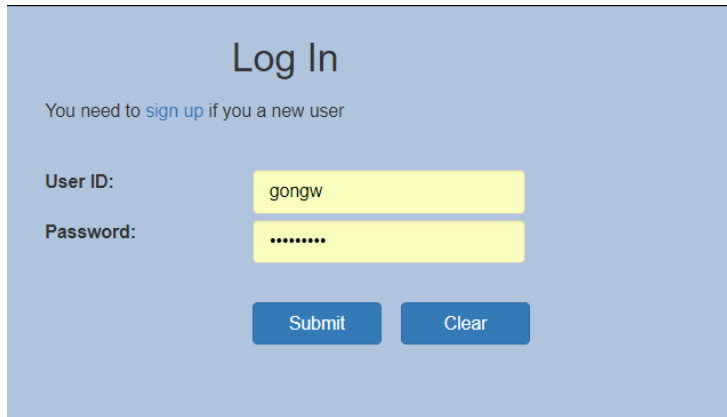
A screenshot of a "Sign Up" form. The title "Sign Up" is at the top. Below it, a note says "All fields are required". The form has five input fields: "User ID:", "Name:", "Phone Number: (nnn-xxx-xxxx)", "Password:", and "Password Again:". At the bottom, there are two buttons: "Submit" and "Clear".

4. Login.php

This page allows the signed-up users to log into the website with their student ID and password.

Except the Landing page (index.php), New User page (NewUser.php) and this page, all other pages of the website are protected, requiring users to be authenticated to access. If an unauthenticated user tries to access a protected page of the website, he/she will be redirected to this login page for authentication.

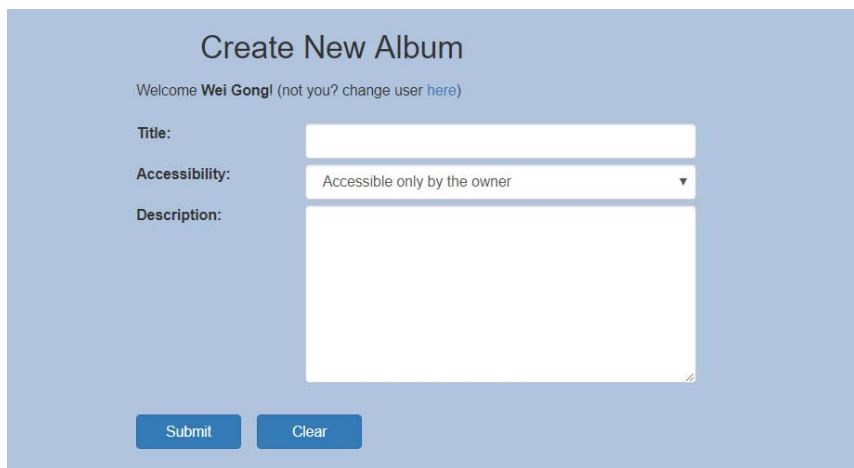
Once successfully authenticated, the user will be redirected **back to the protected page he/she was attempting to access.**



A login form titled "Log In" on a light blue background. Below the title, it says "You need to [sign up](#) if you a new user". There are two input fields: "User ID:" with the text "gongw" and "Password:" with masked characters ".....". Below the fields are two buttons: "Submit" and "Clear".

5. AddAlbum

This page is used to create a new album for the user.



A form titled "Create New Album" on a light blue background. Below the title, it says "Welcome **Wei Gong!** (not you? [change user here](#))". There are three input fields: "Title:" (a text box), "Accessibility:" (a dropdown menu showing "Accessible only by the owner"), and "Description:" (a large text area). Below the fields are two buttons: "Submit" and "Clear".

On this page, the user specifies the title of the album, select the accessibility from the dropdown list and optionally the detailed description of the album. The possible selections of dropdown list come from CST8257 database table **Accessibility**. Currently there are two entries in the table:

- private – The album is only accessible by the user him/her self.
- shared – The album is accessible by the user and the user's friend.

6. MyAlbums.php

This page lists user's albums. For each album, it lists the following information about the album:

- Title – The title of the album
- Date Updated – the last date when the user uploaded the picture(s).

- Number of pictures – The number of pictures the album contains
- Accessibility – the album’s accessibility. It is shown as the selection of the dropdown list. The available selections in the dropdown list come from the database table Accessibility

The user can change the of albums’ accessibility by selecting the desired accessibility from the dropdown list for each album and click the **Save Change** button.

Title	Date Updated	Number of Pictures	Accessibility	
My China Trip	2017-09-04	14	Accessible only by the owner ▼	delete
Massachusetts Institute of Technology	2017-09-04	11	Accessible by the owner and friends ▼	delete
My Australia Trip	2017-09-04	23	Accessible by the owner and friends ▼	delete
Yosemite National Park	2017-09-04	15	Accessible only by the owner ▼	delete
My Europe Trip	2017-09-04	12	Accessible by the owner and friends ▼	delete

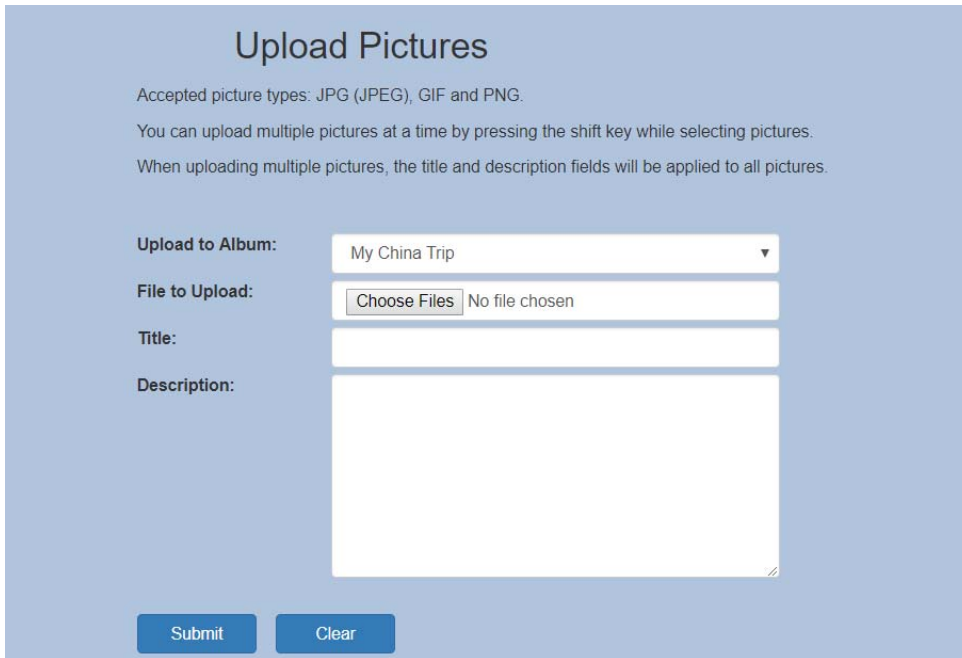
[Save Changes](#)

The page also has following links:

- [Create a New Album](#) – link to **AddAlbum.php** page
- [Album Titles](#) – All album titles in the album list are links to **MyPictures.php** page once clicked, the user is brought to **MyPictures.php** page with the clicked album as the selected album in the dropdown list, see **MyPictures.php** section for details of MyPictures.php page.
- [DELETE](#) – a link button to delete the album. Once clicked, the system will prompt the user to confirm that all pictures in the album will be delete with the album.

7. UploadPictures.php

The user uses this page to upload pictures into one of his/her albums.



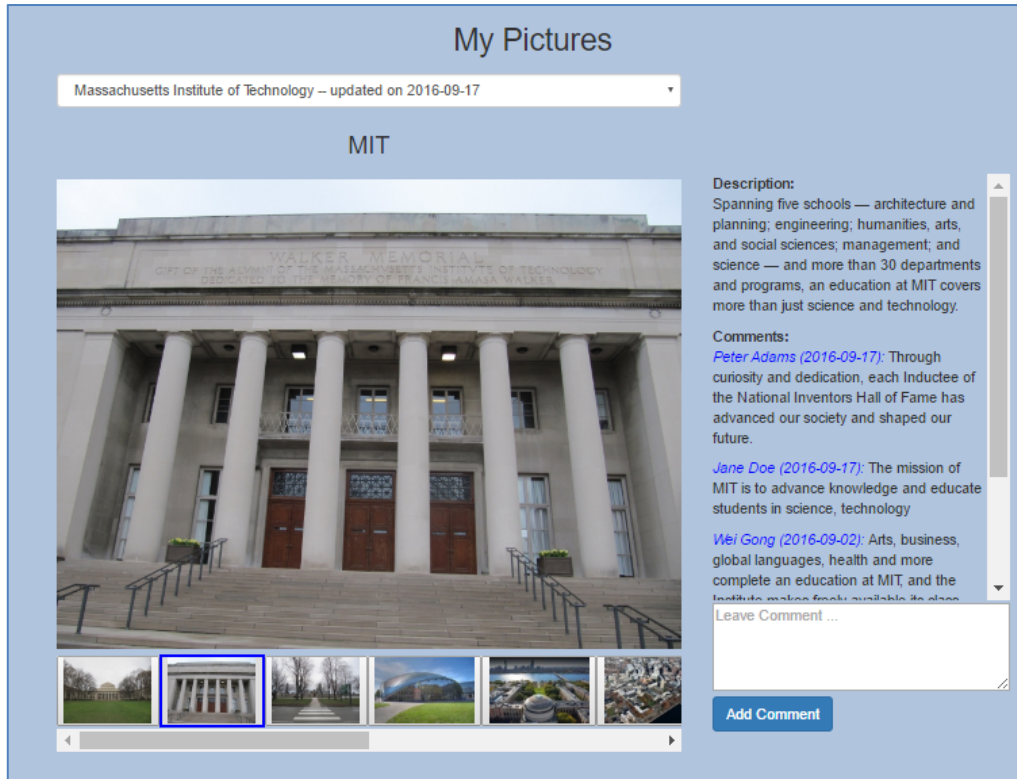
The screenshot shows a web form titled "Upload Pictures" on a light blue background. Below the title, there are three lines of instructional text: "Accepted picture types: JPG (JPEG), GIF and PNG.", "You can upload multiple pictures at a time by pressing the shift key while selecting pictures.", and "When uploading multiple pictures, the title and description fields will be applied to all pictures." The form contains four labeled input fields: "Upload to Album:" with a dropdown menu showing "My China Trip"; "File to Upload:" with a "Choose Files" button and the text "No file chosen"; "Title:" with a single-line text input; and "Description:" with a multi-line text area. At the bottom of the form are two blue buttons: "Submit" and "Clear".

The user can upload single or multiple pictures at a time. Optionally, the user can specify title and description for the pictures to be uploaded. When uploading multiple pictures at a time, the specified title and description apply to all the pictures uploaded.

8. MyPictures.php

This page is for the user to browse and manage his/her pictures. The page contains the following elements:

- A dropdown list for the user to select an album to browse its pictures.
- A picture area showing the picture selected when the user clicks its thumbnail.
- A thumbnail bar displays all thumbnails of the pictures contained in the album. When a thumbnail is clicked the picture area displays the picture the thumbnail represents. The thumbnail of the picture currently in display should be highlighted with a blue border.
- A description and comment area showing the picture's description (if exists) and comments (if any). The comments are ordered chronically from the newest to the oldest.
- A text area for the user to leave a comment. The user can write a comment and click the **Add Comment** button to leave a comment about the picture.



9. AddFriend.php

The user can use this page to send a friend request to another user by enter the other user's user ID and click the **Send Friend Request** button.

A friend request must follow the following rule:

- The entered user ID must exist.
- A user can not send a friend request to her/himself.

- If A sends a friend request to B, while A has a friend request from B waiting for A to accept, A and B become friends.
- If A and B are friends, a friend request between them will result in a message: “You and A (or B) are already friends”.

If the request passes the above rules, the user will get a confirmation message to confirm that the friend request has sent to the specified user:

Add Friend

Welcome **Peter Adams!** (not you? [change user here](#))

Enter the ID of the user you want to be friend with

Your request has sent to Jane Doe (ID: doej).
Once Jane Doe accepts your request, you and Jane Doe
will be friends and be able to view each other's shared
albums.

ID:

10. MyFriends.php

My Friends

Welcome **Wei Gong!** (not you? [change user here](#))

Friends:

Name	Shared Albums	Defriend
John Smith	0	☐
Peter Adams	1	☐
Jane Doe	0	☐

Friend Requests:

Name	Accept or Deny
Mary Johnson	☐

This page lists the user’s friends and friend requests. For each friend, it shows the number of shared album of that friend. The user can perform the following action for each friend.

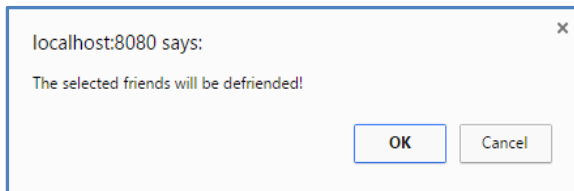
- Click a friend’s name to view the friend’s pictures in the shared albums if that friend has album(s) to share.

- Select the checkbox and click **Defriend Selected** to remove the selected friend(s) from the friend list.

For each friend request, the user can check the checkbox and click **Accept Selected** to accept the friend request to become a friend of the requester. Or click **Deny Selected** to decline the friend request. Once accepted or declined, the friend request will be removed from the friend request list.

The page also contains a link [Add Friends](#) to page **AddFriend.php**

When Defriend a friend or Deny a friend request, the user should be given a proper warning:



11. FriendPictures.php

This page is the same as MyPicture.php except that the album dropdown list only has the friend's shared albums for selection.

